S3.1 The periodic table: Classification of elements

Formation of variable oxidation states in transition elements [HL]

Higher level (HL)

Learning outcomes

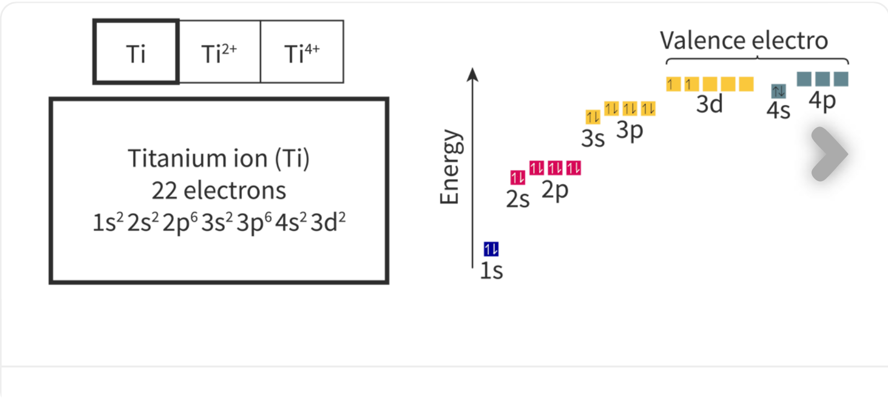
By the end of this section you should be able to:

- Explain why transition elements have variable oxidation states.
- Deduce the electron configurations of ions of the first-row transition elements.

Recall from section S2.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) that when we looked at the cations formed from transition elements, that a variety of different positive charges (oxidation states) were possible. This is why when we named ionic compounds involving transition elements in section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) roman numerals were used to denote which oxidation state the metal was in (i.e. copper(II) sulfate to denote the +2 oxidation state of the copper ion). But why do transition elements form a variety of oxidation states? What is unique about the arrangement of electrons in transition elements that allow the loss of a variable number of electrons?

Electron configuration of ions of transition elements

Transition elements have valence electrons occupying both an s and d-sublevel. Recall from section S2.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) that when these elements lose electrons, they first lose valence electrons from the outer s-sublevel, before continuing to lose electrons from the outer d-sublevel. **Interactive 1** allows you to revisit the electron configurations and sublevels that electrons are lost from for Ti, Ti^{2+} and Ti^{4+} that you studied in section S2.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/).



Interactive 1. Electron Configurations and Orbital Diagrams for Ti, Ti^{2+} and Ti^{4+}.

More information for interactive 1

An interactive slideshow with three slides displays the electronic configurations and orbital diagrams for Ti, Ti^{2+}, and Ti^{4+}.

Slide 1:

Titanium (Ti) has 22 electrons, with an electron configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$. The energy levels increase in the following order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, and 4p. In the orbital diagram, the s orbital has a single box, the p orbital has three boxes, and the d orbital has five boxes. Electrons are represented by arrows inside these boxes. Each box corresponds to a specific orbital in a subshell. The valence orbitals include 3d, 4s, and 4p, and changes in electron occupancy illustrate how Ti loses electrons to form its common ions.

In Ti, 1s, 2s, 3s, and 4s orbitals have two electrons in opposite spins, 2p and 3p orbitals have six electrons, with each orbital containing a maximum of two electrons with opposite spins, and 3d orbital has two electrons in the first two boxes. 4p orbital is empty.

Slide 2:

Titanium (Ti^{2+}) has 20 electrons, with an electron configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$. Two electrons are removed from the 4s orbital of the neutral titanium atom.

Slide 3:

Titanium (Ti^{4+}) has 18 electrons, with an electron configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. Two electrons are first removed from the 4s orbital, and the next two electrons are removed from the 3d orbital, leaving a stable noble gas configuration similar to argon (Ar).
This slideshow helps users understand the electronic configurations of titanium ions, including the orbitals from which electrons are lost.

Worked example 1

Deduce the electron configuration for Fe^{3+}.

When writing the electron configuration for ions, it is helpful to start by writing the electron configuration for the neutral element first.

Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

When iron loses 3 electrons it will first lose 2 electrons from the 4s-sublevel and then the third electron from the 3d-sublevel. This would make its electron configuration:

Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

Worked example 2

Deduce the oxidation state of a vanadium ion with the electron configuration $1s^22s^22p^63s^23p^63d^1$.

A good first step would be to write the electron configuration of a neutral vanadium atom.

V: $1s^22s^22p^63s^23p^64s^23d^3$

Comparing this to the electron configuration in the question we can see the ion has lost two 4s electrons and two 3d electrons for a total of four electrons lost.

Therefore, the oxidation state of the vanadium ion would be +4.

(Note that 4+ would be the ionic charge. Remember that oxidation states and ionic charges have a slightly different notation).

We can look at the successive ionisation energies of transition elements to help explain why they are able to form variable oxidation states. Recall from section S1.3.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/successive-ionisation-energies-hl-id-44045/) that successive ionisation refers to the process of removing successive electrons from an atom or positive ion. **Figure 1** shows the successive ionisation energies for removal of the first five electrons in calcium and vanadium. What do you notice about the energies required to remove successive electrons in vanadium in comparison to calcium?

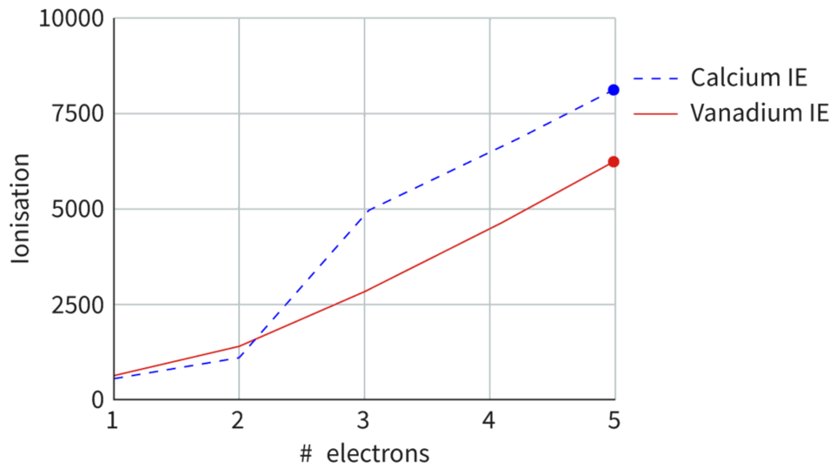


Figure 1. Successive ionisation energies for the first five electrons in calcium and vanadium.

More information for figure 1

The graph displays the successive ionisation energies for the first five electrons of calcium and vanadium. The X-axis represents the number of electrons, ranging from 1 to 5. The Y-axis represents the ionisation energy in arbitrary units, ranging from 0 to 10000.

There are two lines on the graph: 1. A dashed blue line for calcium, showing an increase in ionisation energy, with a marked higher rise after the second electron. 2. A solid red line for vanadium, displaying a more gradual increase in ionisation energies throughout the removal of the five electrons.

Notably, vanadium's ionisation energy increases more gradually than calcium's, which shows a sharp increase after the second electron. This data visually supports the concept of variable oxidation states in vanadium.

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Notice that in vanadium, the ionisation energies required to remove the first five electrons increase at a lower rate when compared to calcium. The successive ionisation energies for calcium show a large increase after the removal of the outer valence shell electrons. Comparing the electron configurations of the two elements in **Figure 2**, why do you think this is? Technically, both elements have only two electrons in the *n* = 4 energy level, so why would vanadium be capable of easily losing more than two?

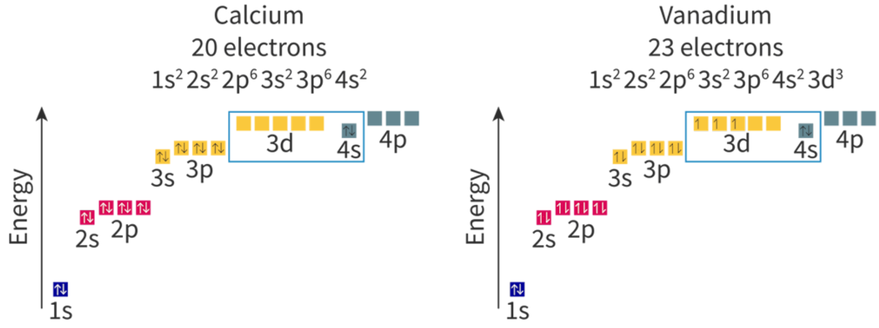

Figure 2. Electron configurations and orbital diagrams for calcium and vanadium.

More information for figure 2

The image depicts two electron configuration diagrams, one for calcium and the other for vanadium, both showing energy levels and orbital occupancy.

On the left is calcium, with the configuration of $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$. The diagram illustrates each orbital with arrows indicating the electrons' spin. The energy levels are arranged vertically, with arrows pointing upwards to show increasing energy. Orbitals are depicted as boxes where paired arrows indicate paired electrons. Notably, the 3d orbitals are empty in calcium.

On the right is vanadium, showing the configuration of $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^3$. Here, similar vertical structure is presented with visible electrons in the 3d orbitals depicted with arrows. This highlights the additional electrons in 3d compared to calcium, explaining vanadium's variable oxidation states and easier ionization from higher energy levels.

The overall layout emphasizes the energy differences and orbital filling in these two elements, relevant to their chemical properties.

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Since the 4s and 3d-sublevels are close in energy, electrons from both of these sublevels are able to act as valence electrons. Electrons are lost first from the 4s-sublevel; explaining why many transition metal ions form the +2 oxidation state. However, electrons can also easily be lost from the 3d-sublevel without applying much additional energy for removal. This is why transition elements can easily form ions with variable oxidation states as there are a number of d electrons that can be lost in addition to s electrons. **Figure 3** shows the common oxidation states of the metals from groups 3–12 in period 4.

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
								+1	
	+2	+2	+2	+2	+2	+2	+2	+2	+2
+3	+3	+3	+3	+3	+3	+3			
	+4	+4		+4					
		+5							
			+6	+6					
				+7					

Figure 3. Common oxidation states for period 4 metals in groups 3— 12.

More information for figure 3

The image is a table displaying common oxidation states for period 4 metals in groups 3 through 12. The horizontal axis lists the metals: Sc (Scandium), Ti (Titanium), V (Vanadium), Cr (Chromium), Mn (Manganese), Fe (Iron), Co (Cobalt), Ni (Nickel), Cu (Copper), and Zn (Zinc). The vertical axis represents the oxidation states: +1, +2, +3, +4, +5, +6, and +7.

- Sc shows the oxidation state of +3.
- Ti shows +2, +3, and +4.
- V shows +2, +3, +4, and +5.
- Cr shows +2, +3, +4, +5, and +6.
- Mn shows +2, +3, +4, +6, and +7.
- Fe shows +2 and +3.
- Co shows +2 and +3.
- Ni shows +2.
- Cu shows +1 and +2.
- Zn shows +2.

The table illustrates that, except for Zn, most metals can have multiple oxidation states, reflecting the variable oxidation capabilities of transition metals, primarily due to 3d sublevel electron interactions.

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Looking at **Figure 3** note that zinc only shows the +2 oxidation state. This is another reason to support the exclusion of zinc from transition elements. You may also notice that scandium is listed as having the +3 oxidation state. There are active disputes in the scientific community about whether scandium should be considered a transition element. Some argue it shouldn't be included because only the +3 oxidation state is common. Others argue it should be included because it is possible to form the +2 oxidation state, it is just uncommon to encounter.

Aspect: Science as a shared endeavour

Disputes in science are not uncommon. Can you think of ways that disputes between scientists help the production of knowledge in chemistry? Can you think of ways that disputes hinder the production of knowledge in chemistry? If an element does not fit neatly into a category that we, as humans, have created, should we exclude it or consider it an exception?

Figure 4 shows all the successive ionisation energies for vanadium, labelling the sublevels where electrons were lost from. Notice that the 4s and 3d electrons have successive ionisation energies close together, suggesting that they all behave as valence electrons. This also acts as further evidence that the 4s and 3d-sublevels are close in energy.

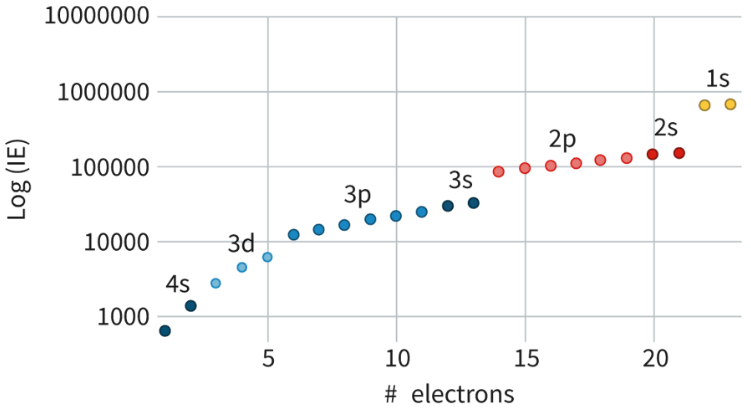

Figure 4. Successive ionisation energies for vanadium.

More information for figure 4

The image is a graph illustrating the successive ionisation energies for vanadium. The X-axis represents the number of electrons, labeled from 0 to 25. The Y-axis is the logarithm of the ionisation energy (Log (IE)) and spans from 1,000 to 10,000,000 in logarithmic scale. Data points on the graph represent different sublevels from which electrons are removed.

The sublevels are labeled as 4s, 3d, 3p, 3s, 2p, 2s, and 1s, with each sublevel's electrons identified by colored dots (blue, red, yellow). The graph shows that the 4s and 3d sublevels are close in energy as their ionisation energies are grouped closely together, supporting the idea that they behave similarly as valence electrons. The ionisation energy increases steadily with electrons removed, and a distinct rise in ionisation energy can be noted as transitions to 3p and 3s occur, with further sharp increases at the 2p, 2s, and finally the 1s sublevels, indicating a higher energy required to remove core electrons.

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Knowledge of the oxidation number for transition elements is very important since they can lead to very different properties. Titanium(IV) oxide, for example, is utilised as an active ingredient in sunscreen because it reflects a high percentage of UV radiation, whereas titanium(II) and titanium(III) oxides do not. **Table 1** shows a comparison of these titanium oxides. Note that titanium does not have a +5 oxidation state as this would require the removal of an electron from the 3p sublevel.

Table 1. Titanium oxides with variable oxidation states and their properties.

Substance name	Chemical formula	Oxidation number of titanium	Properties
Titanium(II) oxide	TiO	+2	Brown solid. Melts at approximately 175
Titanium(III) oxide	Ti_2O_3	+3	Black, semiconducting solid. Melts a approximately 2100 °C
Titanium(IV) oxide	TiO_2	+4	White solid. Melts at approximately 180 Reflects UV radiation and is therefore us sunscreen.

Activity

- **IB learner profile attribute:** Risk-taker
- **Approaches to learning:** Social skills — Appreciating the diverse talents and needs of others
- **Time required to complete activity:** 20 minutes
- **Activity type:** Group activity

Successive ionisation energy paper telephone

In groups of 5. Each person needs a stack with 5 paper squares. No talking with one another until all rounds have been completed. Each round will involve you writing/sketching on the top square and then passing the entire stack clockwise to the next person. When you receive a stack of squares, look only at the top square, move it to the bottom and complete the next round on the new square on top.

- Round 1 — select a transition element from period 4. Write it on the top sheet and pass it clockwise.
- Round 2 — Read the top square, move it to the bottom. Write the electron configuration of the neutral element on the top square.
- Round 3 — Read the top square, move it to the bottom. Sketch the plot of successive ionisation energies for this element based on its

electron configuration. Sketch on the top sheet and pass clockwise when finished.

- Round 4 — Look at the sketch on the top square, move to the bottom. Write down the electron configuration for the element having a +3 oxidation state. Pass clockwise when finished.
- Round 5 — Read the top square, move it to the bottom. Write the name of the element. Pass clockwise when finished.

You should now have your original stack. Look at the element written. Is it the same one you started with? One by one go through your entire stack with your group. If the elements at the beginning and end do not match as a group discuss where the misalignment occurred.